Response to Reviewers

Predicting Verification Latency for Post-Quantum Archival Evidence

We thank the editor and reviewers for their careful reading and constructive comments. We have revised version 1 to address the requested clarifications. Each response below identifies the change and its location in version 2. The original experimental results and conclusions are retained. The public code and data link is now given under Reviewer 2 and in the availability statements.

Reviewer 1

Comment 1. Generalization across hardware platforms, cryptographic libraries and storage environments.

We agree that the original discussion did not sufficiently distinguish transfer of the method from transfer of its coefficients. Section 5.2 now makes that distinction explicit. It describes local workload instrumentation, independent count reconciliation, refitting, rank and residual checks and evaluation on fresh histories after freezing the fit. Hardware, library, parser and storage changes can alter the relative costs of the terms. The present evidence therefore supports scale extrapolation within one environment; it does not establish unchanged accuracy across platforms or production storage systems.

Comment 2. Calibration overhead and practical deployment requirements.

Thank you for requesting this practical clarification. Section 3.4.1 now distinguishes measurement collection, coefficient fitting and prediction. It identifies the 432 balanced observations, 36 medians and 36×7 fitting matrix used by the seven-term model. Calibration-only collection time, solver time and solver peak memory were not separately reported, so we do not provide an unsupported runtime estimate. We also describe prediction arithmetic and the verifier instrumentation, timing, solver and environment records needed for deployment.

Comment 3. Trusted key enrollment, simulated timestamps and long-term archival trust.

We agree and have expanded Section 3.1.1. The enrollment policy is supplied by a trusted caller; checking agreement with that policy does not establish that the enrollment process was honest. Signed modeled times authenticate asserted values and do not supply trusted timestamp evidence. RFC 3161 is cited for the trustworthy-time requirement. We distinguish successful implementation checks from the broader assurances needed for archival trust, including acquisition, key custody, validation evidence and timely renewal. We also state that distinct signature families do not eliminate shared enrollment or administrative failures and that an eligible surviving branch cannot repair compromised source evidence.

Comment 4. Presentation of model coefficients and adaptation to other evidence systems.

We have revised Table 4 to identify each coefficient symbol, the work it counts and its value in seconds and microseconds. The accompanying explanation distinguishes fixed overhead from per-unit terms and clarifies that the jointly fitted coefficients are empirical allocations. A worked example applies the existing coefficients to an E full query with 100,000 records and five renewals and reproduces the prediction already reported in Table 6. Section 5.2 explains how practitioners can identify suitable workload terms, refit locally and validate the model on fresh histories.

Comment 5. Cold-cache verification and network retrieval as future evaluation scenarios.

We have added a brief discussion in Section 5.3 covering cold local retrieval and remote storage access. It identifies the need for explicit cache-state reporting, end-to-end timing, component spans and network conditions. We also explain that overlapping stages must be accounted for and that restarting a worker does not establish cold caches. These scenarios are presented as future work.

Reviewer 2

Comment 1. Single-environment evaluation and absence of independent replication.

We agree. Section 5.1 retains the single-host and independent-replication limitations stated in version 1. Sections 5.2 and 5.3 now clarify their practical implications for adapting the model to other platforms and storage environments. We do not claim cross-platform validation or independent replication.

Comment 2. Public availability of code and datasets.

We agree that public access would strengthen reproducibility. We have prepared VITAL_Q_GitHub_Full_Reproducibility.zip, a repository package with the original frozen code, observations, coefficients, analysis, retained selective proofs, executed notebook and plotting code. Within its vital-q folder, data/published/ preserves the original inputs and results, src/ preserves the original scientific source; scripts/ provides the portable execution workflow. Numerical replay matched 2,092 reported numerical fields. The README distinguishes replay of the original results from fresh measurement collection and provides setup and execution instructions.

A separate full-scale execution on 12 October 2026 completed 72 development and 90 evaluation histories, yielding 864 and 1,080 measurements. All 233 qualification checks passed and all 486 newly retained selective proofs were reverified. This rerun used locally refitted coefficients and produced median absolute relative errors of 2.12% in balanced mode and 2.21% in isolated mode; both modes satisfied the original support rule. These are separate reproducibility results, not replacements for the manuscript's 1.63% and 2.55%, coefficients, tables or figure. They do not establish cross-platform validation or independent replication.

The fresh run is retained in executions/full_scale_20261012/. Its scope, portable execution amendments and an excluded initial cleanup failure are documented in validation/FULL_SCALE_EXECUTION.md. This rerun used local checkpoints; the manuscript's hosted-snapshot statement refers to the original experiment. The retention limitation in Section 5.1 and Appendix A.4 remains: full archives and full-query proofs were verified during collection and discarded, while selective proofs were retained. The code supports regeneration; new timings and signature bytes need not be identical.

The package is now public at https://github.com/ discoversaikatchakraborty/VITAL-Q. Release v1 is the reproducibility package, release v2 adds the revised manuscript together with this response, release v3(https://doi.org/10.5281/zenodo.23329917) adds the DOI . The data availability and code availability statements cite that location.

Reviewer 3

Acceptance assessment. Measurement corrections and methodological contribution.

We thank the reviewer for the encouraging assessment and acceptance recommendation in the supplied excerpt. We have retained the explanation of the measurement correction and qualification checks in Sections 3.2 and 5 and Appendix A.4. The manuscript also retains the qualification that the observed improvement cannot be attributed to the timer correction alone, because the model, fitting procedure and measurement protocol changed together. No additional change was requested in the excerpt.

We appreciate the opportunity to improve the manuscript. The manuscript revisions remain limited to the reviewers' requests and preserve the original study. The separate execution is documented in the reproducibility package. Reviewer 2's public-availability request is met by the public repository cited in the availability statements.

Summary of where each comment is addressed

Comment	Issue	Addressed in
R1 Comment 1	Generalization	Section 5.2; limitations in Section 5.1
R1 Comment 2	Calibration and deployment	Section 3.4.1
R1 Comment 3	Security assumptions and archival trust	Section 3.1.1; Section 5.1; Appendix A.2; references [1] and [12]
R1 Comment 4	Coefficients and adaptation	Table 4; Sections 4.3 and 5.2; Equation (1)
R1 Comment 5	Cold-cache and network scenarios	Section 5.3
R2 Comment 1	Evaluation limitations	Sections 5.1–5.3; conclusion; Appendix A.4
R2 Comment 2	Public code and data	Availability statements; reproducibility package README and execution report; https://github.com/discoversaikatchakraborty/VITAL-Q
R3 Assessment	Acceptance acknowledgment	Response above; retained Sections 3.2 and 5; Appendix A.4